Harsha Madhamanchi

Software Engineer

Java • Spring Boot • AWS • Distributed Systems • DevOps • GenAI

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Professional Summary

Senior Software Engineer with 7 years of experience building distributed backend systems, cloud-native applications, and engineering platforms. Strong expertise in Java, Spring Boot, and AWS, with hands-on experience in event-driven architectures, production reliability, and CI/CD automation. Improved SaaS tenant onboarding time by 40% and reduced infrastructure release timelines by 30% through automation..

Professional Experience

Software Engineer II
May 2024 – October 2026

FICO (Fair Isaac Corporation)

- Co-designed and implemented a five-service event-driven distributed system on AWS ECS using Java 21 and Spring Boot, leveraging SQS FIFO queues to coordinate case routing between fraud decision engines and external communications platforms.
- Implemented fault-tolerance mechanisms using Resilience4j, incorporating Circuit Breaker and exponential backoff retry patterns to handle downstream service failures, prevent cascading failures, and improve microservice resilience.
- Provisioned multi-component AWS infrastructure (ECS Fargate, DynamoDB, DocumentDB, Lambda, S3) using CloudFormation IaC across dev/stage/prod environments. Implemented changeset-based deployment workflows and automated Jenkins CI/CD pipelines, reducing release timelines by 30%.
- Authored Architectural Decision Records (ADRs) for production MySQL encryption-at-rest and database upgrade strategies, adopted across cross-team architectural review forums.
- Built a Business Knowledge Extraction system using LLaMA 3 (8B) and LangChain RAG pipelines, exposed through an MCP client/server layer integrated into the existing microservice ecosystem to automate report summarization.
- Developed API-driven production incident alerting integrated with Microsoft Teams, automatically notifying relevant on-call engineers to improve incident visibility and response coordination.

Senior DevOps Engineer
April 2023 – May 2024

Philips Innovation Campus

- Managed AWS infrastructure across 5 environments using Terraform automation.
- Implemented one-click infrastructure provisioning workflows and mentored junior DevOps engineers, improving deployment efficiency through Terragrunt adoption.
- Led the migration of mission-critical CI/CD pipelines from Azure DevOps to GitHub Actions, standardizing automated release workflows and achieving 100% migration success with zero production defects.

Senior Software Engineer
April 2022 – April 2023

Philips Innovation Campus

- Drove early customer adoption of the SaaS Factory Platform by proactively engaging with initial customers, understanding their challenges, and collaborating on platform-based solutions to improve adoption and organizational reach.
- Delivered Tenant Onboarding as one of the three major EPICs within the SaaS Factory Platform, leveraging Spring Boot and AWS Lambda to improve onboarding time by 40%.
- Contributed to architectural design decisions and developed a stateless provisioning component integrated with ServiceNow (ITSM) to automate tenant-specific resource creation, enabling customers across different subscription models to onboard and access applications.
- Integrated core services with AWS S3, SQS, and RDS to support scalable data processing.

Software Engineer *August 2020 – March 2022*

Philips Innovation Campus

- Played a key role in building Nethra, an internal engineering intelligence platform, as part of the founding three-member engineering team.
- Developed Python-based, cron-driven integrations with Jira and Azure Boards to centralize project timelines, feature tracking, sprint progress, and delivery forecasting.
- Integrated Black Duck, SonarQube, and Jenkins to consolidate security, code quality, and build health metrics, providing engineering visibility across multiple Philips business verticals.
- Developed and maintained backend services using Python and Spring Boot, alongside Terraform-based AWS infrastructure automation, supporting internal engineering workflows and consistent provisioning across environments.

Software Engineering Intern *August 2019 – August 2020*

Philips Innovation Campus

- Developed Selenium-based test automation scripts to automate application testing and improve regression testing efficiency.
- Contributed to the MEL (Monitoring and Evaluation) project by developing frontend components using Angular, supporting application functionality and UI development.

Technical Skills

Programming Languages: Java, Python, TypeScript
Frameworks & Libraries: Spring Boot, Angular, LangChain
Cloud & DevOps: AWS (ECS, Lambda, S3, SQS, RDS, DynamoDB, DocumentDB), Docker, Terraform, Terragrunt, Jenkins, GitHub Actions, CI/CD, Kubernetes
Databases: PostgreSQL, MySQL, DynamoDB, DocumentDB
AI & GenAI: MCP, Retrieval-Augmented Generation (RAG), LLaMA 3, LangChain
Architecture & Engineering: Microservices, Event-Driven Architecture, Distributed Systems, REST APIs, System Design, Infrastructure as Code

Key Achievements

- Delivered a major Tenant Onboarding EPIC within the SaaS Factory Platform, reducing tenant onboarding time by 40
- Developed an AI-powered Business Knowledge Extraction System using LLaMA 3 and RAG, earning internal recognition for innovation.
- Supported zero-downtime deployments through high-availability AWS infrastructure and automated multi-environment deployment pipelines.
- Led a team to victory in an internal innovation hackathon, competing against 40+ teams.

Education

Vellore Institute of Technology (VIT), India *2020*
M.Tech in Integrated Software Engineering **8.85 CGPA**